PSYLIQ Project 1: Scam Execution & Victim Profiling

PSYLIQ Investigative Data Analysis Internship
Project 1: Scam Execution & Victim Profiling — Final Report
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PSYLIQ Investigative Data Analysis Internship

Project 1: Scam Execution & Victim Profiling — Final Report

Prepared for: Psyliq Data Analysis Internship Review **Analysis period covered:** 2020–2024 **Primary sources:** FBI Internet Crime Complaint Center (IC3) Annual Reports; FTC Consumer Sentinel Network (CSN) Data Books; U.S. Department of Justice (DOJ) press releases and case materials; CISA public advisories (supporting)

Executive Summary

This report analyzes five years (2020–2024) of U.S. scam and fraud data to answer three questions: **who** is being victimized and **where**, **how** scammers execute successful fraud, and **how** government enforcement is responding. Complaint volume rose from roughly 792,000 in 2020 to 860,000 in 2024, while reported losses nearly quadrupled, from $4.2B to $16.6B, indicating that the average scam is becoming substantially more costly even where complaint counts are flat.

The single most consequential finding of this analysis is a **enforcement-to-victimization gap**: phishing/spoofing is the largest driver of complaint volume in 2024 (193,407 complaints), yet none of the ten major DOJ enforcement actions reviewed for this project targeted phishing directly. Enforcement instead clusters around elder-care fraud, Medicare fraud, and data-broker-facilitated mail fraud — important categories, but not the categories driving the largest share of victim contact.

Methodology & Sources

Data was compiled from three official public sources:

- **FBI IC3 Annual Reports (2020–2024)** — victim demographics (age group, state), scam/crime type

breakdowns, and reported loss figures.
- **FTC Consumer Sentinel Network Data Books (2020–2024)** — payment method demands and consumer-reported fraud contact methods.
- **U.S. Department of Justice press releases and case materials (2020–2024)** — ten major anti-scam enforcement actions, including agency, targeted scam type, and recovery amount.

All data was compiled into a structured Excel workbook (`Data_Matrix_P1.xlsx`), validated and analyzed in Python (`Psyliq_P1_Scam_Profiling.ipynb`), including a SQLite-based relational query for the government-response analysis.

Data cleaning notes: - Two `Government_Actions` fields contained mixed data types in the source material (one date recorded only as a month, one recovery amount recorded as "not specified in a report"). These were parsed into a clean numeric/date format and explicitly flagged rather than estimated. - IC3's phishing-family category name changed across report years (`Phishing/Vishing/Smishing/Pharming` in 2020–2021, `Phishing` in 2022, `Phishing/Spoofing` in 2023–2024). Because this analysis uses single-year (2024) snapshots for scam-type ranking, this naming shift does not distort the reported findings, but it is noted here for transparency.

Section 1.1 — Victim Demographics & Geographic Hotspots

Age Group Trends (2020–2024)

Year	Age 20–29	Age 60+
2020	70,791	105,301
2021	69,390	92,371
2022	57,978	88,262
2023	62,410	101,068
2024	71,399	147,127

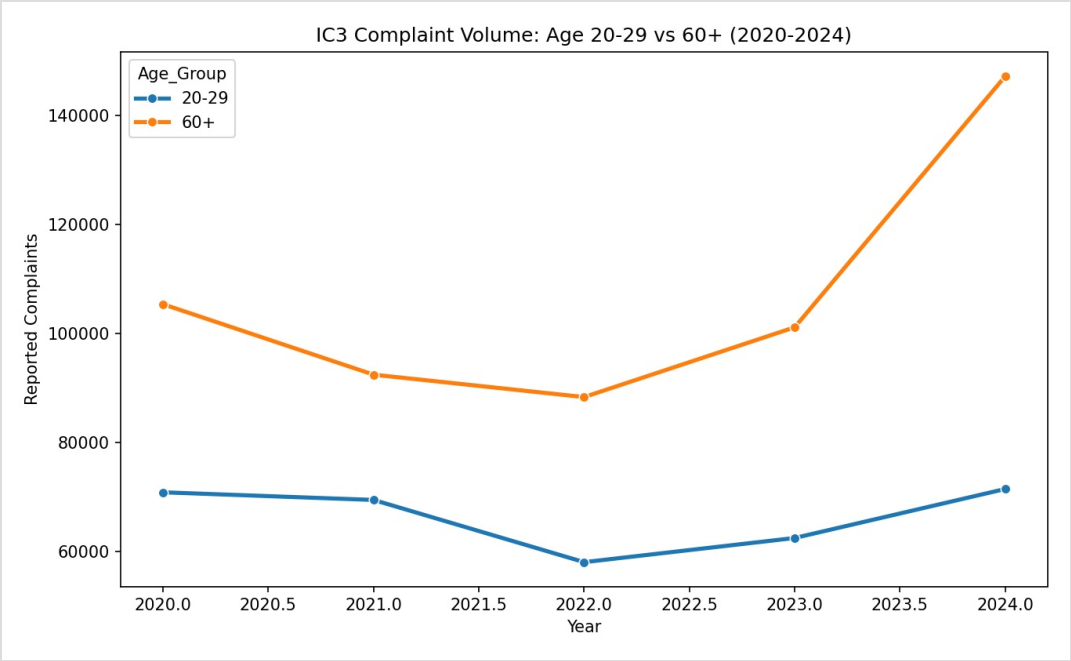
Between 2020 and 2024, complaints from the 20–29 age group grew just 0.9%, essentially flat. Complaints from the 60+ age group grew **39.7%** over the same period, with a particularly sharp jump in 2024 alone (up from 101,068 to 147,127 — a 45.6% single-year increase). The `Scam_Script` data attributes much of this to "pig butchering" style crypto-investment scams and emergency/relative-in-crisis framing, both of which disproportionately target older victims building long-term trust with a scammer.

Top 10 States by Complaint Volume (2024)

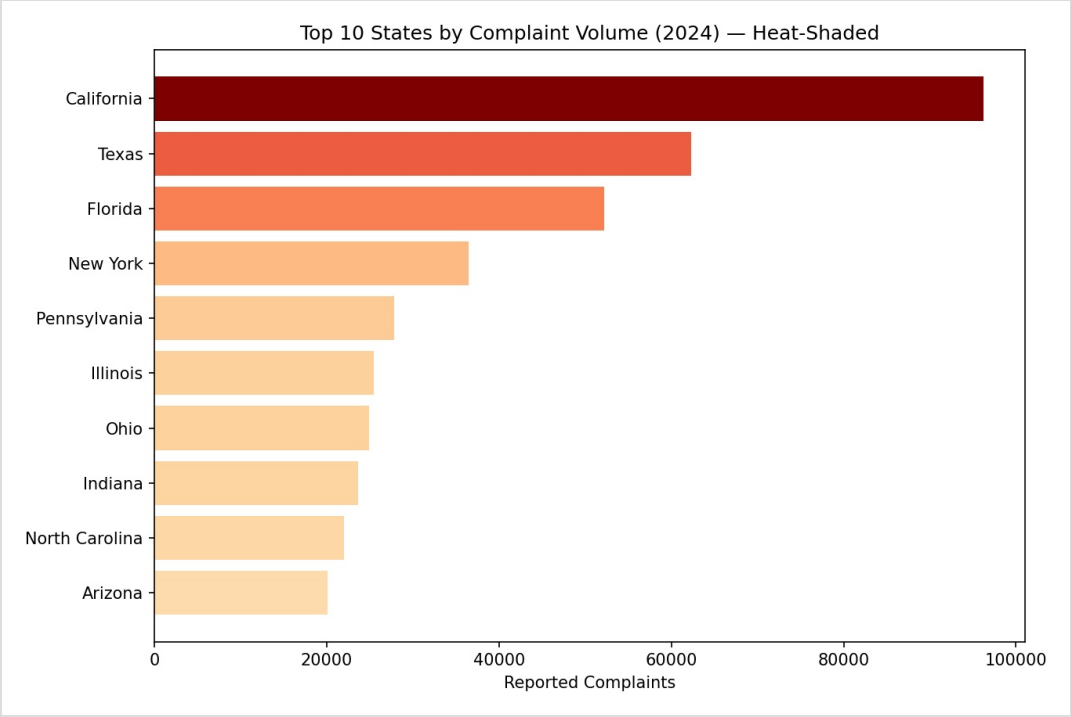
Rank	State	Complaints	Reported Loss
1	California	96,265	$2.54B
2	Texas	62,347	$1.35B
3	Florida	52,191	$1.07B
4	New York	36,468	$904M
5	Pennsylvania	27,838	$400M
6	Illinois	25,446	$479M
7	Ohio	24,915	$278M
8	Indiana	23,659	$125M
9	North Carolina	22,021	$324M
10	Arizona	20,101	$392M

The top 4 states by raw volume are the four most populous U.S. states, suggesting complaint counts track population size closely at the top of the distribution — loss-per-complaint is a more useful comparative metric than raw volume alone (e.g., Personal Data Breach cases averaged over $22,000 in loss per complaint

nationally in 2024, far above the overall average).



Complaint volume by age group, 2020-2024



Top 10 states by complaint volume, 2024

Observation Task 1.1

The most unexpected finding is the acceleration, not just the existence, of growth in the 60+ demographic. A 39.7% five-year increase would be notable on its own, but nearly all of that growth happened in a single year (2023→2024), coinciding with the rise of cryptocurrency-based “pig butchering” investment scams. This suggests the 60+ vulnerability trend is not a slow demographic drift but a rapid response to a specific new scam technique — meaning intervention strategies need to move as fast as the scam methodology does, not on an annual review cycle.

Section 1.2 — Scammer Tactics & Intervention Analysis

Top 5 Scam Types by Complaint Volume (2024)

Scam Type	Complaints	Reported Loss
Phishing/Spoofing	193,407	$70.0M
Extortion	86,415	$143.2M
Personal Data Breach	64,882	$1.45B
Non-Payment/Non-Delivery	49,572	$785.4M
Investment	47,919	$6.57B

Note the inversion between columns: Phishing/Spoofing dominates by complaint *count* but ranks lowest of these five by dollar *loss*, while Investment fraud is the fifth-most-reported but by far the costliest — nearly $6.57B from under 48,000 complaints, an average loss well over $130,000 per case. This distinction matters for intervention design: volume-based scams need broad, low-cost consumer education, while investment fraud needs fewer but much higher-touch interventions.

Top Payment Methods Demanded (2024)

Payment Method	Complaints
Credit Cards	108,881
Payment App or Service	90,571
Debit Card	76,285

Payment Method Shift, 2020 → 2024

2020's top 3 (in order): Credit Cards, Debit Card, Payment App or Service. 2024's top 3 (in order): Credit Cards, Payment App or Service, Debit Card.

Payment App or Service (e.g., Zelle, Venmo, Cash App) overtook Debit Card between 2020 and 2024. This reflects a broader shift toward instant, hard-to-reverse payment rails — payment apps offer scammers speed and finality that traditional card transactions (which carry chargeback protections) do not.

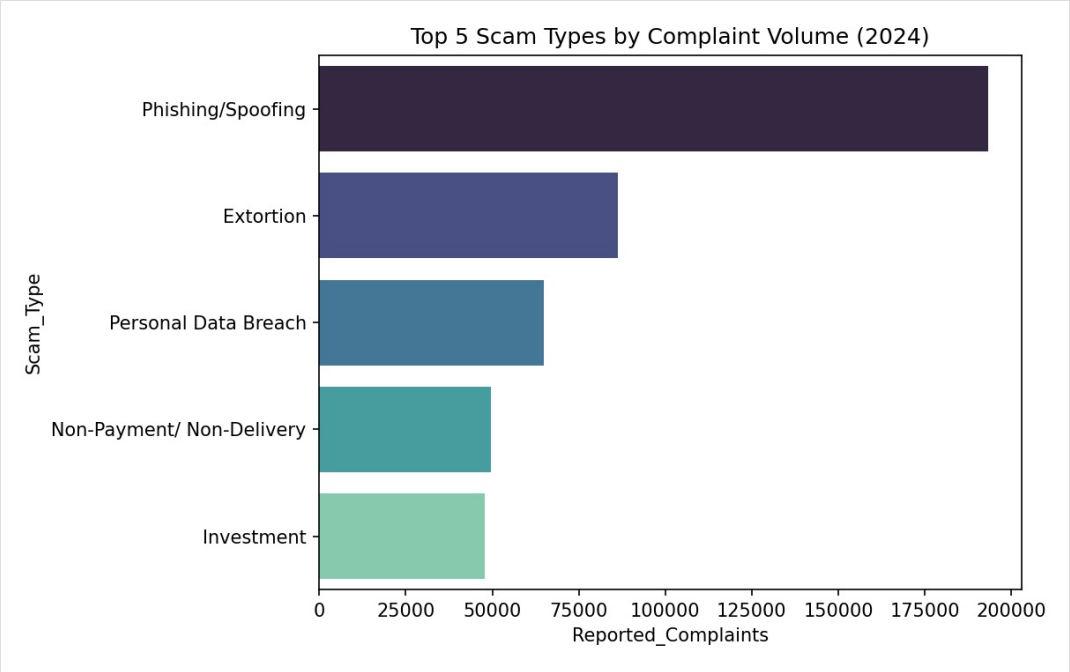
The Scammer Script (2024 Pattern, per IC3 data)

Based on the `Scam_Script` and `Scam_Tactics` data, a successful imposter/investment fraud in 2024 typically follows six stages:

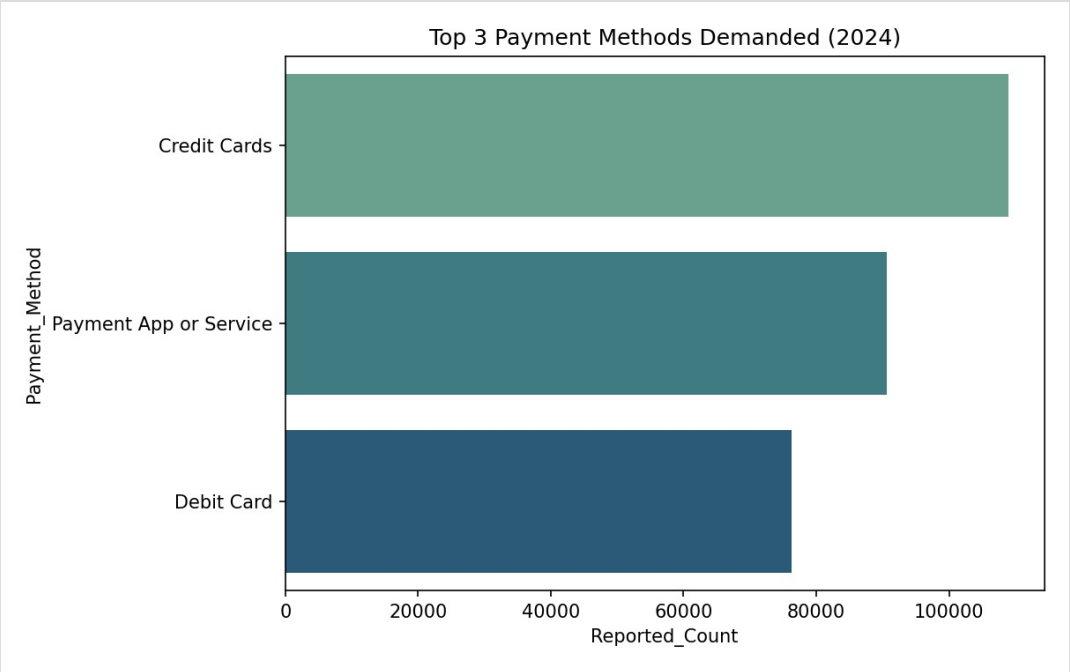
1. **Initiate contact** — via smishing (toll-debt texts), phishing/spoofing messages, or cold outreach on dating/social platforms.
2. **Build trust** — impersonating a cryptocurrency exchange, government agency, real estate attorney, or a slow-building romantic relationship.
3. **Create urgency or fear** — "relative in crisis," closing-deadline pressure on a real estate wire, or a fabricated account compromise alert.
4. **Direct victim to act** — instructing withdrawal of cash, purchase of gold for courier pickup, or deposit into a cryptocurrency ATM.
5. **Extract funds** — overwhelmingly via cryptocurrency in 2024 (149,686 complaints referencing crypto, $9.3B in associated losses — the single largest loss vector of any method tracked).
6. **Move and launder funds** — rapid transfer to overseas accounts, commonly in Hong Kong, Vietnam, Mexico, the Philippines, India, and China, before any recovery attempt can begin.

Top Urgency-Language Themes

The most frequent recurring urgency themes in the qualitative tactics data were: **threat** (of data destruction/exposure), **pressure** (time-limited offers or compliance demands), and **urgency** framing tied to account compromise or benefit access. Fifteen of the forty-four logged tactic entries did not specify explicit urgency language in the source material — a data availability limitation, not an indication that urgency tactics were absent.



Top 5 scam types by complaint volume, 2024



Top 3 payment methods demanded, 2024

Observation Task 1.2

The clearest shift is from card-based to app/crypto-based payment demands, and this signals a shift in scammer infrastructure toward irreversibility, not just convenience. Credit and debit cards carry built-in chargeback and fraud-dispute mechanisms; payment apps and cryptocurrency generally do not. The rise of cryptocurrency as the single largest loss vector ($9.3B in 2024) indicates scam operations have professionalized around payment rails specifically chosen to defeat recovery — this is an infrastructure choice, not an incidental trend, and it should inform any real-time intervention strategy Psyliq builds.

Section 1.3 — Government Response & Future Defense

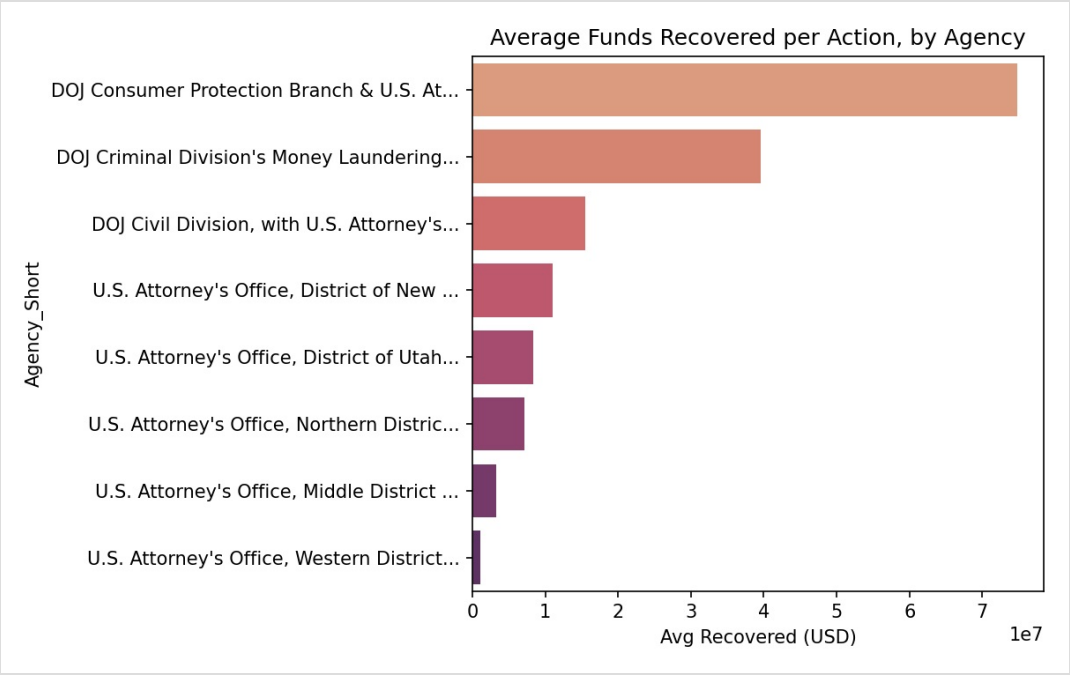
Government Enforcement Actions Analyzed (10 Actions, 2020–2024)

A relational table of ten major DOJ actions was built and queried using SQLite. The query below answers the brief's required question:

```sql
SELECT Agency, COUNT(*) AS num_actions, AVG(Est_Funds_Recovered_Clean) AS avg_recovered
FROM actions
WHERE Est_Funds_Recovered_Clean IS NOT NULL
GROUP BY Agency
ORDER BY avg_recovered DESC
```

Answer: The **DOJ Consumer Protection Branch, jointly with the U.S. Attorney's Office for the District of Colorado**, has the highest average estimated funds recovered per action among the ten cases analyzed — averaging **$74,750,000 per action** across its two logged cases (the Epsilon Data Management data-broker settlement and a related mail-fraud victim-compensation case), for a combined $149.5M recovered. This substantially outpaces the next-highest agency grouping, reflecting the outsized recovery available in large-scale civil settlement and victim-compensation actions relative to individual criminal restitution orders.

One of the ten logged actions (`P1-A003` , U.S. v. Shawn Phillips) had no numeric recovery figure in the source material — the available reporting only stated loss amounts, not recovered funds — and was therefore correctly excluded from this average rather than estimated.



Average funds recovered per action, by agency

The Future of Anti-Scamming

Based on this research, three concrete ways Psyliq could apply emerging technology to stay ahead of these threats:

1. **Real-time payment-rail risk scoring.** Since cryptocurrency and payment apps now drive the largest share of unrecoverable losses, Psyliq could build transaction-pattern models that flag high-risk transfer requests (e.g., a first-time crypto ATM deposit following urgent contact) before funds leave the account, rather than after the fact.
2. **Cross-referencing enforcement gaps against real-time complaint trends.** This project's own gap finding — phishing dominates complaints but is nearly absent from major DOJ actions — suggests real value

in a live dashboard that flags when a rapidly growing scam category has no corresponding enforcement action, so resources can be redirected faster than the current annual reporting cycle allows.
3. **Natural-language pattern detection on scam scripts.** The six-stage script pattern identified in this report (contact → trust → urgency → directed action → extraction → laundering) is consistent enough across scam types that NLP-based message screening — flagging urgency-plus-payment-instruction language in real time — could intercept fraud at the "direct victim to act" stage, before funds are extracted.

Observation Task 1.3

The clearest enforcement gap is the mismatch between where complaint volume is concentrated and where major DOJ actions are targeted. Phishing/spoofing drove the largest share of 2024 complaints (193,407), yet zero of the ten logged government actions in this dataset targeted phishing directly — enforcement instead concentrated on elder-care fraud, Medicare fraud, and data-broker-facilitated schemes. This is not necessarily a full picture of DOJ's total enforcement activity (this analysis reviewed ten representative actions, not DOJ's complete caseload), but within this sample, it suggests a private-sector solution focused specifically on phishing-vector detection and interception would be addressing a category that is comparatively underserved by major public enforcement actions.

Data Limitations

- **DOJ recovery and date figures:** One of ten logged government actions had no stated recovery amount in the source reporting and was excluded from recovery averages rather than estimated. One date was recorded only to month-level precision.
- **IC3 category naming:** The "phishing" category name changed across report years (2020–2021 vs. 2022 vs. 2023–2024); this report uses single-year (2024) comparisons to avoid distortion, but any multi-year phishing trend analysis would need to account for this naming change first.
- **Ten-action sample size:** The government-response analysis is based on ten representative DOJ actions, not DOJ's complete 2020–2024 caseload, and should be read as an illustrative sample rather than an exhaustive enforcement audit.
- **IC3 vs. FTC terminology:** IC3 complaint/report counts are not equivalent to unique victims (one victim may file multiple reports); FTC Consumer Sentinel figures represent consumer reports, which may include reports filed by third parties on a consumer's behalf.

Conclusion

Across all three modules, this analysis points to one connected story: fraud is shifting toward higher-value, harder-to-reverse payment methods (cryptocurrency, payment apps), disproportionately targeting older victims through patient, trust-building scripts, while major public enforcement action is concentrated in different categories than where complaint volume is highest. A private-sector intervention that closes the specific gap identified here — real-time detection focused on the phishing-to-crypto pipeline — would address the point in the scam lifecycle where public enforcement, based on this sample, currently has the least direct presence.